Movie Data Analysis Using Python

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Subject: Python Programming / Data Analysis
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INTRODUCTION

This project performs data analysis on a movie dataset using Python libraries such as Pandas, NumPy, Matplotlib and Seaborn. The goal of the project is to analyze movie ratings, views and genres to generate useful insights and visualizations.

INSTALL PACKAGES

1) pip install numpy
2) pip install pandas
3) pip install matplotlib
4) pip install seaborn

CODE IMPLEMENTATION

```
import numpy as np

import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns



# Load the DataSet

df = pd.read_csv(r"C:\Users\DHRUV SHAH\Python\project 3\MovieData.csv")

print(df.head(10))



# Data inspection

print(df.info())

print(df.describe())
```

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```
# Data Cleaning

print(df.isnull().sum())

df['Year'] = df['Year'].astype(int)

df['Views'] = df['Views'].astype(int)

print(df.duplicated().sum())

print(df.info())


# Basic analysis

avg_movieRating = df['Rating'].mean()

max_HighestRating = df.loc[df['Rating'].idxmax(),'Rating']

min_LowestRating = df.loc[df['Rating'].idxmin(),'Rating']

HighestViews = df.loc[df['Views'].idxmax(),'Title']

HighestGenre = df.loc[df['Genre'].idxmax(),'Rating']

HighestYear = df.loc[df['Year'].idxmax(),'Rating']



print(f"Average Movie Rating is {avg_movieRating}\n")

print(f"Max Rating movie is {max_HighestRating}\n")

print(f"Min Rating movie is {min_LowestRating}\n")

print(f"Highest View is {HighestViews}\n")

print(f"Highest Genre is {HighestGenre}\n")

print(f"Highest Year is {HighestYear}\n")


# Advance Analysis

most_Viewed = df.nlargest(5,'Views')[['Title','Views']]

print(most_Viewed)
```

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```
most_Rated = df.nlargest(5,'Rating')[['Title','Rating']]

print(most_Rated)



# Data visualization

sns.barplot(x="Genre", y="Rating", data=df, color="y")

plt.show()


sns.displot(df['Rating'], kde=True, color="r")

plt.show()


sns.scatterplot(x="Views", y="Rating", data=df, color="b")

plt.show()


sns.lineplot(x="Year", y="Rating", data = df, color="r")

plt.show()


sns.countplot(x="Genre",data=df, color="g", hue="Title")

plt.show()


# Average views per Genre

avg_views = df.groupby('Genre')['Views'].mean()

print(avg_views)


most_watchedGenre = df.loc[df['Genre'].idxmax(),'Genre']

print(f"Most Watched genre is {most_watchedGenre}")


# Find number of movies Relased each year
```

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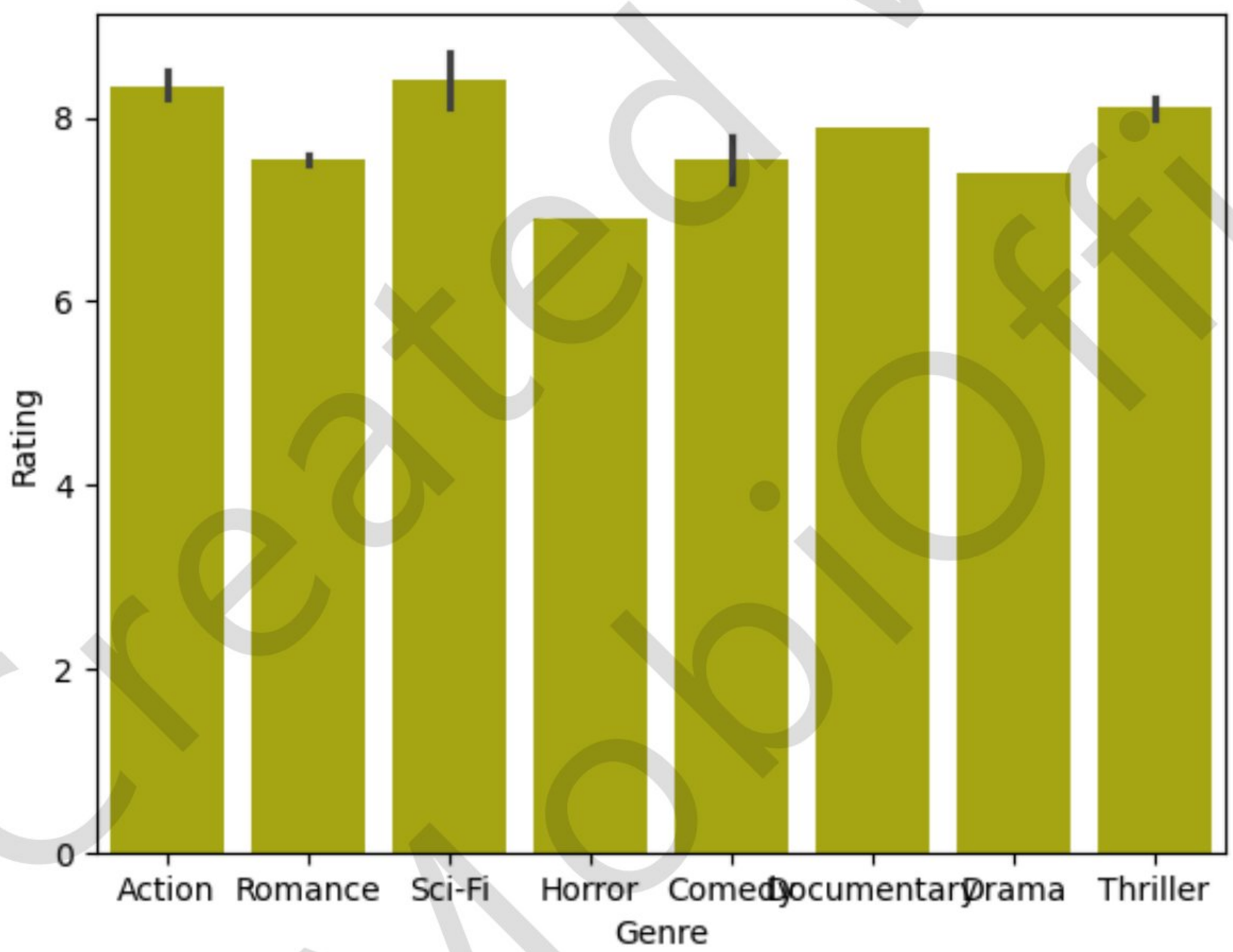
```
sns.barplot(x="Title", y="Year", data=df, color="y")

plt.show()



# Find Median rating of movies

Median_Rating = df['Rating'].median()

print(Median_Rating)
```



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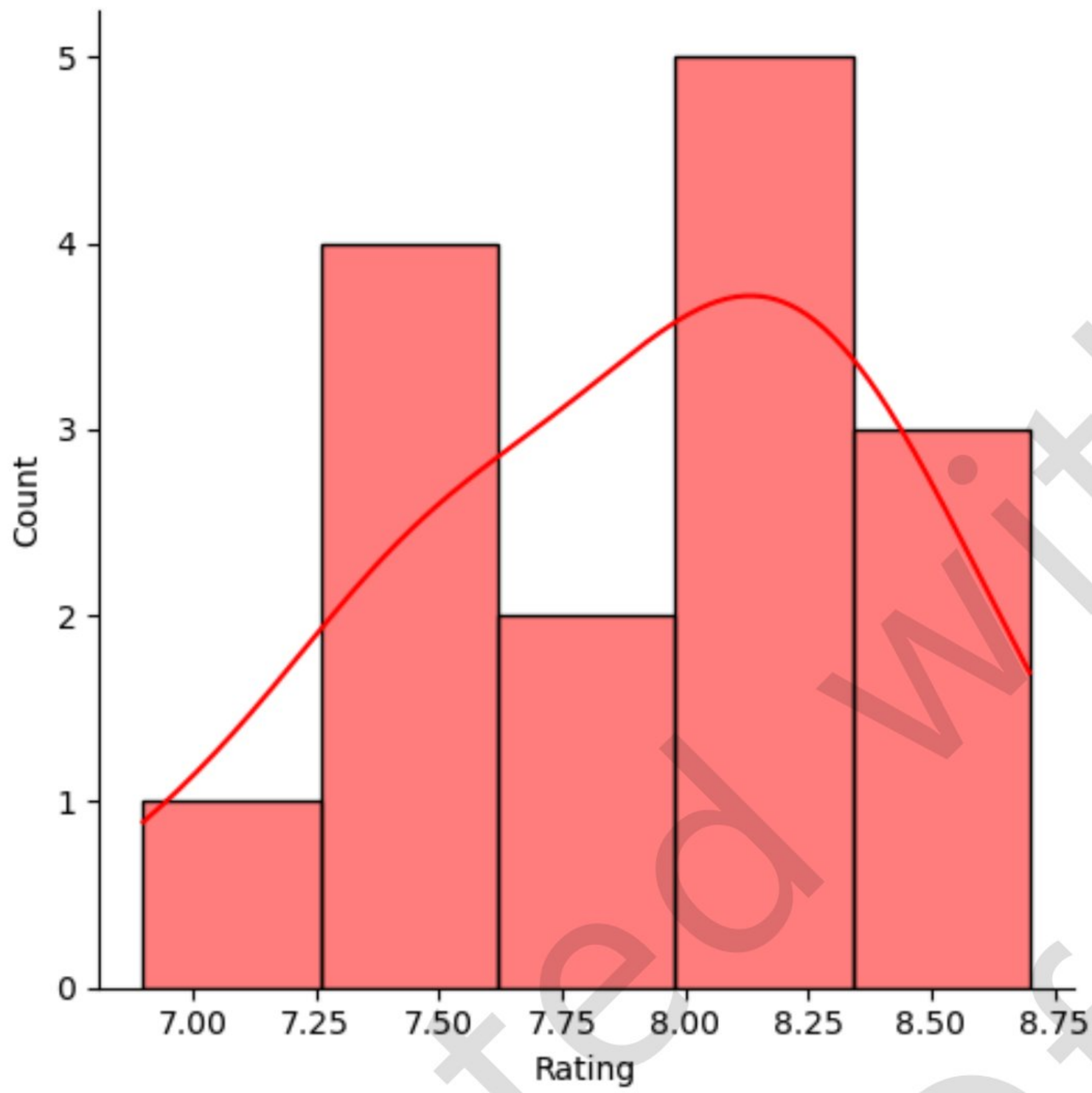

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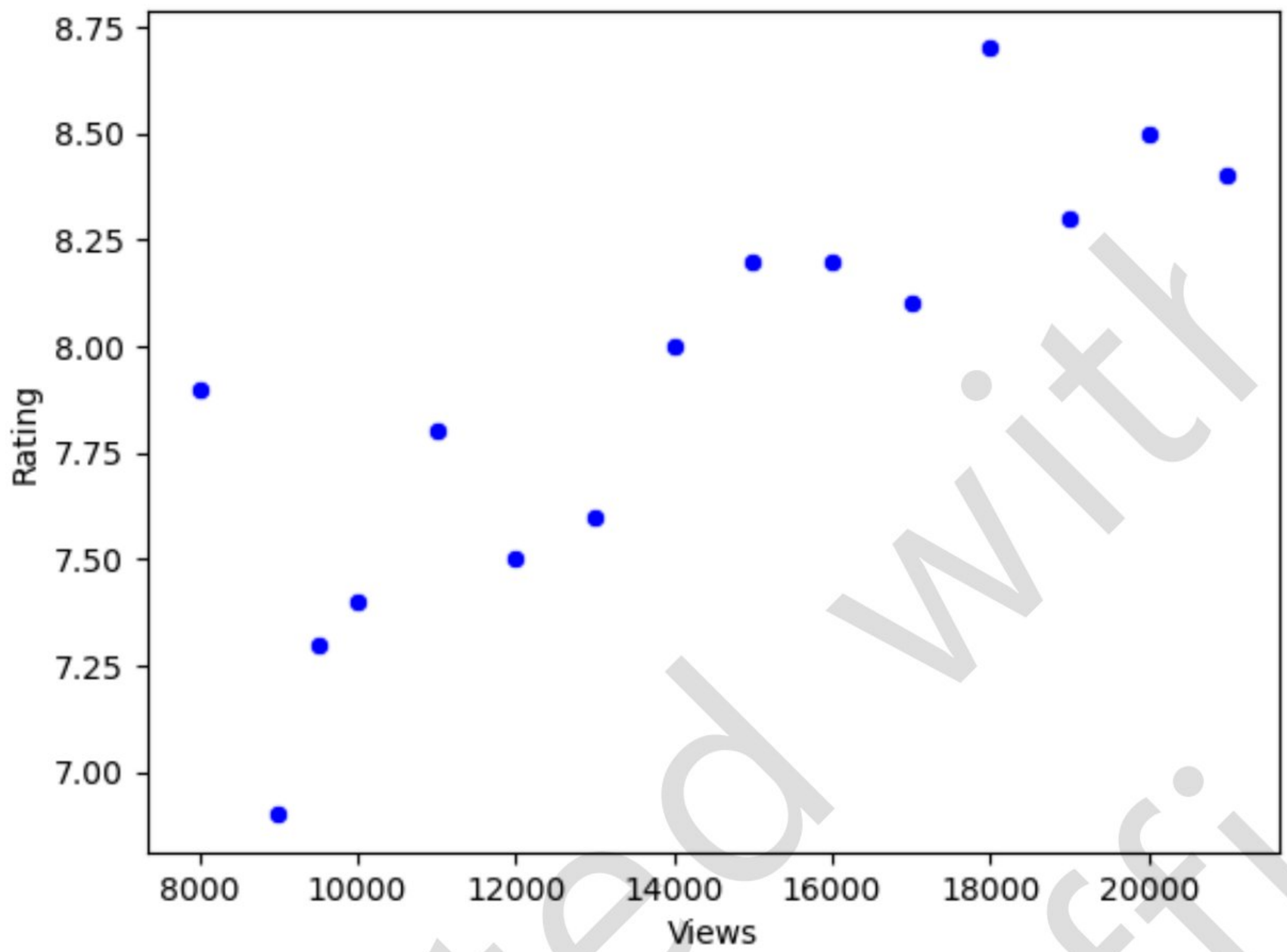

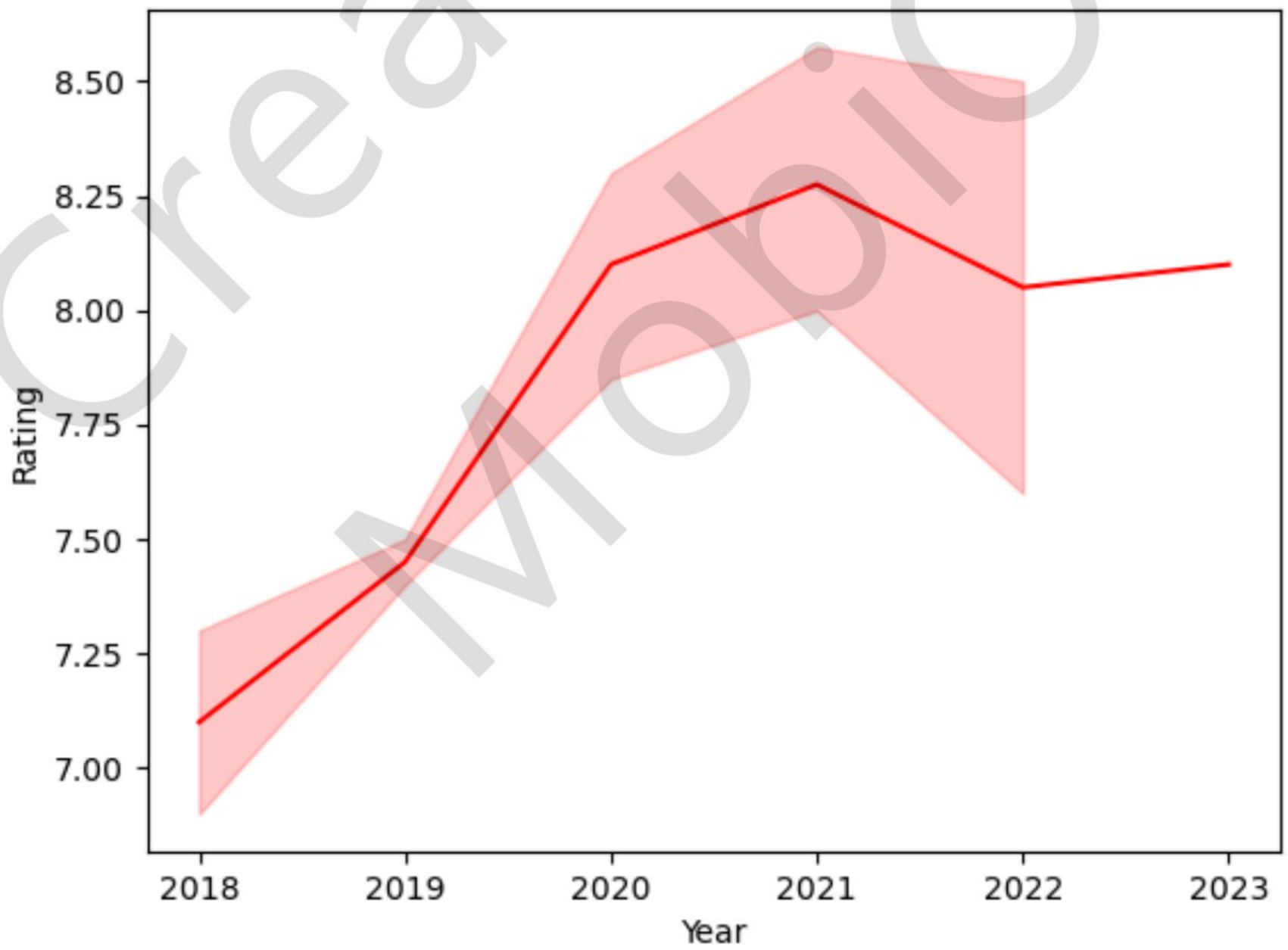

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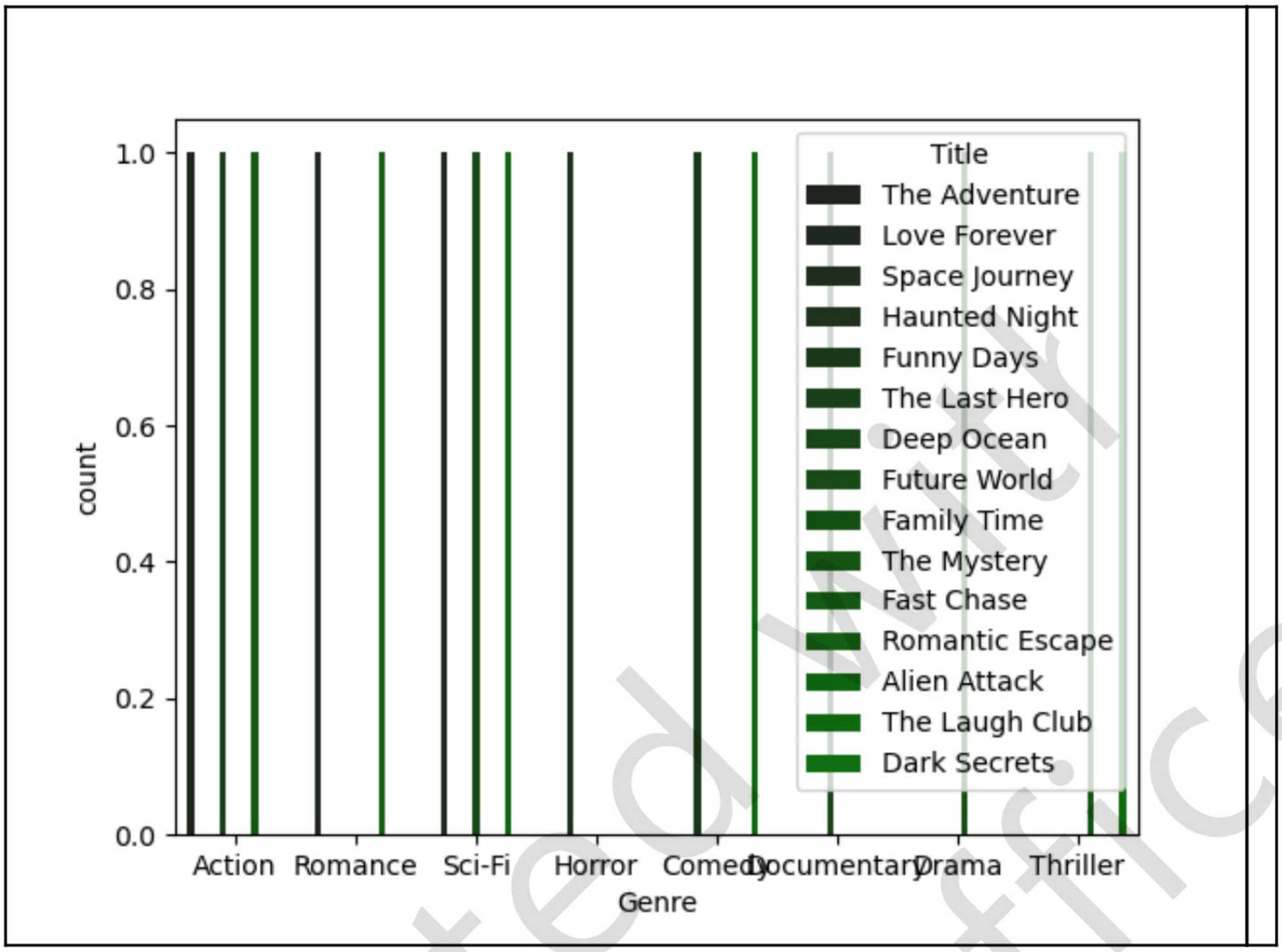

OBSERVATIONS

1) Action and Sci-Fi movie have higher average ratings compared to other genres.

2) Recent movies (2021-2023) have higher average ratings compared to old movies.

3) Some genres like Horror or Comedy have lower average ratings.

4) A small number of movies are extremely popular.

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